LESSON PLAN

AFCS (22)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Block diagram of Guidance and control loop
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Block diagram of Guidance and control loop.
5. Specific Objectives	22.1.1.1Trainee will be able to understand the basic of guided weapon systems and types of targets.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of block diagram components; Safety: No physical handling of live systems; Supervision: Instructor monitors diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a guidance system?
(b) Name two types of guided weapon systems.
(c) Define the term 'control loop' in the context of missiles.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Guidance and Control Loop • Input Signal Generation • Error Detection Mechanism • Actuator Response • Feedback Path Components	L & Ds	PPT, IB	15 min
(b)	• Signal Processing Unit Functions • Servo Motor Operation • Stabilization Systems • Target Acquisition Logic • Control Surface Deflection	L & Ds	PPT, PDF	20 min
(c)	• Closed Loop vs Open Loop Systems • Gain Adjustment Parameters • System Stability Criteria • Noise Filtering Techniques • Loop Closure Verification	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the feedback signal corrects the missile's trajectory.
(b) Describe the role of the actuator in the control loop.

12. Home Assignment and Follow Up	Prepare a brief note on the agencies for achieving missile guidance to be discussed in the next lesson plan (LP 03).

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Agencies for achieving missile guidance and classifications of missile guidance system.	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min